LUNDS UNIVERSITET

Numerical Analysis for E3

Numerical Analysis, Centre for Mathematical Sciences

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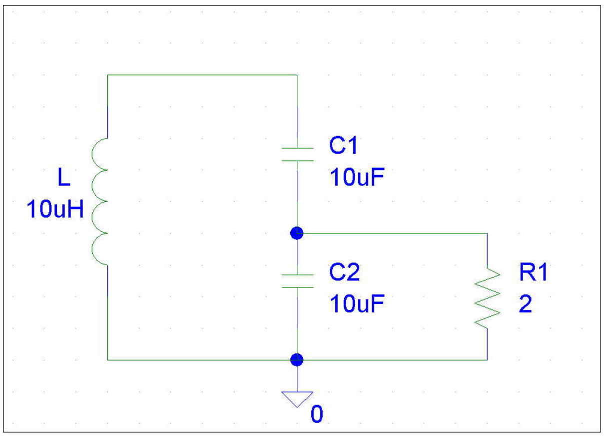
Homework assignment 1

Delivery date: March 25th 2003, no later than 18:00h

Aims of this assignment: Getting started with MATLAB; Polynomial interpolation.

1. A simple circuit. In the introduction to this course An LC-circuit was modelled by differential equations. For example The circuit depicted in the figure,



is described by

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}V_{C_1} &= -\frac{i_L}{C_1} \\
\frac{\mathrm{d}}{\mathrm{d}t}V_{C_2} &= -\frac{i_L}{C_2} - \frac{V_{c2}}{RC_2} \\
\frac{\mathrm{d}}{\mathrm{d}t}i_L &= \frac{V_{C_1} + V_{C_2}}{L}
\end{aligned}
\tag{1}
$$

Task 1a): Implement the right hand side of system (1) as a MATLAB–function. Let $L = C_1 = C_2 = 10^{-5}$ and $R = 2, 100, 10K$ and compare your results.

Task 1b): Simulate this circuit by solving the initial value problem. Realistic initial conditions are $V_{C_1}(0) = 1, V_{C_2}(0) = 1$ and $i_L(0) = 0$. Apply the MATLAB–solver `ode45`. Simulate for the time interval $[0, 10^{-3}]$.

2. Interpolation of tabulated data. The following table shows the Dow Jones industrial index in 2001

month	1	2	3	4	5
index	10600	11000	10400	9800	10800

month	6	7	8	9	10
index	11000	10500	10500	10000	8800

The data is taken the first day in each month.

Task 1a): Apply Lagrange interpolation to find the ploynomial of lowest possible degree interpolating the data from July to September. Do this computation “by hand”. Determine the three relevant Lagrange polynomials to this data and compose them to find the interpolation polynomial. Evaluate the polynomial to estimate the index in April and in January. If you compare this result with the given data (in April and January), what is your conclusion?

Task 1b): Compute the interpolating polynomial to all the data using MATLAB functions `polyfit` and `polyval`. Plot the interpolating polynomial from January to December. Can you comment the figure?

3. Lagrange Polynomials. Prove that Lagrange functions are linearly independent.

Hint: Show that if $\sum_{i=0}^{n} a_i L_i^n(t) = 0$ then it follows that all coefficients $a_i = 0$.